

James K. Mwaura

Nairobi, Kenya · +254 722 922 852 · mwaura.kimani@gmail.com ·

GitHub: github.com/jamesmwaura007

Professional Summary

Data-driven professional with hands-on experience in analytics, machine learning, and data visualization. Skilled in transforming raw data into actionable insights that drive business decisions. Strong foundation in Python, SQL, dashboarding, and predictive analytics. Adept at identifying trends, automating workflows, and developing models that improve operational efficiency and business performance.

Technical Skills

- **Python:** NumPy, Pandas, Scikit-learn, OOP, Data Structures, Django, Flask
 - **Machine Learning:** Supervised & Unsupervised Learning (Decision Trees, Random Forests, SVM, KNN, Naive Bayes, K-Means), TensorFlow/Keras
 - **Deep Learning:** CNN, RNN, LSTM, Transformers
 - **Databases:** SQL, MySQL, PostgreSQL
 - **MLOps:** Model deployment, monitoring, CI/CD
 - **Visualization:** Tableau, Power BI, Data Studio, Plotly Dash
 - **Other:** Web scraping, Git version control
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Education

Moringa School – Data Science

Comprehensive training in Python, Machine Learning, Deep Learning, SQL, Visualization, and MLOps.

BSc. Information Technology

(JKUAT)

Focus: Networking, Databases, Systems Development, Programming

Data Science Projects

Customer Churn Prediction – SyriaTel

Built ML models (Logistic Regression, Random Forest) to predict churn; Random Forest improved accuracy and highlighted key churn drivers (e.g., customer service calls, day charges). Delivered business recommendations for proactive retention.

Movie Production Analysis

Analyzed IMDB & Box Office datasets to identify profitable genres, revenue trends, and rating correlations. Produced insights guiding investment decisions and marketing strategies.

Aviation Safety & Accident Analysis

Processed 80k+ aviation accident records, cleaned missing data, and performed risk profiling. Identified low-risk aircraft (e.g., Boeing 737-800, Airbus A300) and critical safety factors such as weather conditions and engine configuration.

MovieLens Recommendation System

Developed collaborative and hybrid filtering models using the 100k MovieLens dataset. Cleaned and analyzed 220k+ records, addressed cold-start issues, and generated personalized Top-5 recommendations per user.

Credit Card Fraud Detection (XGBoost & Random Forest)

Analyzed 440k+ transactions with severe class imbalance. Applied SMOTE, feature engineering, and advanced ML models. Achieved high AUC-ROC (>0.93) using XGBoost, generating actionable insights for fraud-risk mitigation.

Experience Highlights

Data & Business Insights (Various Roles)

- Analyzed large datasets to identify performance trends and support business decisions.
 - Automated recurring reports reducing manual processing time by 40%.
 - Created dashboards for leadership teams to monitor KPIs, customer behavior, and market performance.
 - Presented insights to cross-functional teams to support strategic initiatives.
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Certifications

- Data Science / Machine Learning Certification
 - Google Data Analytics / IBM Data Science
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Core Strengths

Analytical Thinking · Problem Solving · Stakeholder Communication · Attention to Detail · Continuous Learning